

Last week : Classical Integrability

PCM & Sine-Gordon

Spectral curve

This week : Semiclassical Quantization

Scattering

Superstring on $AdS_5 \times S^5$

§ Semiclassical Quantization

Quantize classical integrable systems

- historically how QISM was found

(Covariant) quantization of $AdS_5 \times S^5$ is hard

- no 'proof' of quantum integrability

PCM $\leadsto \{L, L\} = \delta'$: non-ultralocal

$\rightarrow$ also hard!

PCM Lagrangian:

$$\mathcal{L} = \frac{1}{2} G_{MN} \partial_\alpha X^M \partial^\alpha X^N = \frac{1}{2} \text{tr} (g^{-1} \partial_\alpha g \ g^{-1} \partial^\alpha g)$$

polar coordinates in S^3 : $(q^1, q^2, q^3) = (\theta, \phi, \psi)$

$$ds^2 = d\theta^2 + d\phi^2 + d\psi^2 + 2 \cos \theta \ d\phi \ d\psi$$

$$\rightarrow p_i = \frac{\delta \mathcal{L}}{\delta \partial_0 q^i}, \quad \{q^i(\sigma), p_j(\sigma')\} = \delta^i_j \delta(\sigma - \sigma')$$

Want to compute the Poisson bracket

$$\{j_\alpha(\sigma), j_\beta(\sigma')\} = \dots$$

We can derive Poisson brackets in many ways,
but they are invariant under canonical transform

$$\{q^i, p_j\} \rightarrow \{Q^i, P_j\}$$

If we use polar coordinates,

$$\hat{j}_\alpha = g^{-1} \partial_\alpha g = \underline{T^a} \underline{E^a_i} \partial_\alpha q^i$$

$SU(2)$

generators

vierbein of S^3

$j_0^a \sim E^a_i \partial_0 q^i$: momentum, $E^a_i q^i$: "coordinate"

Eventually we obtain

$$\{j_1^a(\sigma), j_1^b(\sigma')\} = 0$$

(no momentum)

$$\{j_0^a(\sigma), j_0^b(\sigma')\} = -f^{abc} j_0^c \delta(\sigma - \sigma')$$

(Poisson-Lie)

$$\{j_0^a(\sigma), j_1^b(\sigma')\} = -f^{abc} j_1^c \delta(\sigma - \sigma') - \underline{f^{ab} \delta'(\sigma - \sigma')}$$

non-ultralocal term

Lax connection of PCM:

from $j_1 \sim g^{-1} \partial_1 g$

$$J_\sigma(\sigma, \tau, x) = \frac{j_0 - x j_1}{1 - x^2}$$

Poisson bracket is

[Maillet, Dorey-Vicedo 0606202]

$$\{J_{\sigma}(\sigma, \tau, x), J_{\sigma'}(\sigma', \tau, x')\}$$

$$= [r(x, x'), J_{\sigma} \otimes 1 + 1 \otimes J_{\sigma'}] \delta(\sigma - \sigma')$$

$$- [s(x, x'), J_{\sigma} \otimes 1 + 1 \otimes J_{\sigma'}] \delta(\sigma - \sigma')$$

$$- 2 s(x, x') \delta'(\sigma - \sigma') \quad \text{still non-ultralocal}$$

r : antisymmetric, $\hbar \rightarrow 0$ limit of R -matrix

$$R = I + \frac{\hbar}{u} P \equiv I + \hbar r$$

s : symmetric $\leadsto$ ambiguity in $P_{\exp}$, Jacobi id, etc.

Maillet introduced "Symmetric brackets" to show

$$\{ \text{tr} \Omega(x), \text{tr} \Omega(x') \} = 0$$

But how to quantize PCM remains unclear

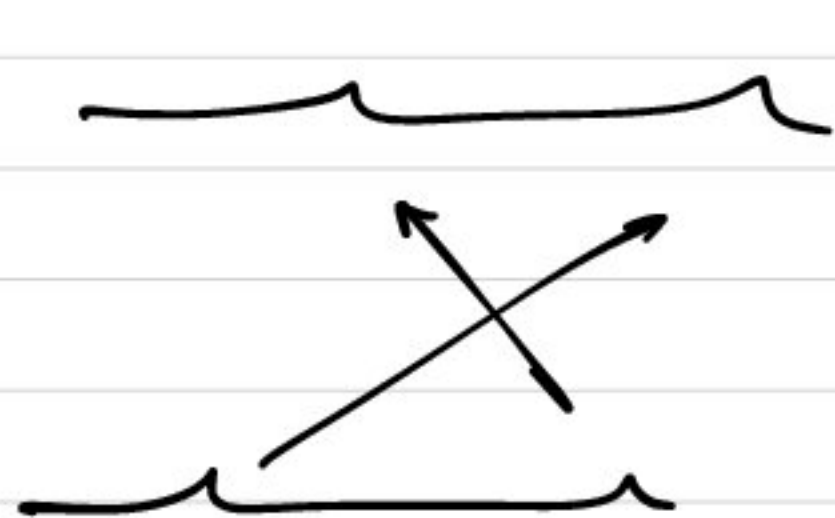
$\Rightarrow$ needs different method to find R-matrix

In sine-Gordon, there is a gauge transformation

which removes $\partial_x \phi$ from Lax connection

- Poisson brackets are ultralocal
- Use QISM to find R-matrix
- (inverse) Pohlmeyer doesn't solve PCM problem

§ Scattering of giant magnons [Hofman, Maldacena]



Compute classical S-matrix from
soliton scattering

Compare with "Berke Ansatz" like
prediction (AFS photo)

This is worldsheet scattering

~ wave-function for dilatation eigenstates

Not spacetime scattering ~ 4pt fn of AdS/CFT

N giant magnons in $\mathbb{R}_t \times S^3$

$\updownarrow$ Pohlmeyer

N soliton solutions in (complex) sine-Gordon

$\uparrow$

Dressing method ; for $N=2$

$$\left\{ \begin{array}{l} \phi = 4 \arctan \left(\frac{a_1 + a_2}{a_1 - a_2} \frac{e^{\xi_1} - e^{\xi_2}}{1 + e^{\xi_1 + \xi_2}} \right) \\ \partial_x \partial_T \phi = \sin \phi, \quad \xi_i = a_i X + \frac{T}{a_i} \end{array} \right.$$

Lax connections in lightcone coordinate:

$$L_T = i\lambda\sigma_3 + \frac{i\partial_T\phi}{2}\sigma_1, \quad L_X = \frac{-i}{4\lambda} e^{\frac{i\phi}{2}\sigma_1} \sigma_3 e^{-\frac{i\phi}{2}\sigma_1}$$

$$(\partial_\alpha - L_\alpha)\psi = 0 \quad \text{for } \alpha = X, T$$

The vacuum solution: $\phi = 0$

$$(\partial_X - i\lambda\sigma_3)\psi_0 = \left(\partial_T + \frac{i}{4\lambda}\sigma_3\right)\psi_0 = 0$$

$$\Rightarrow \psi_0 = \exp\left[\left(i\lambda X - \frac{iT}{4\lambda}\right)\sigma_3\right]$$

Dressing transform :

$$\psi = g(\lambda) \psi_0, \quad g(\lambda) = 1 + \sum_{i=1}^N \frac{R_i}{\lambda - \lambda_i}$$

$$0 = (\partial_\alpha - L_\alpha) \psi_0 = g(\partial_\alpha - L_\alpha) g^{-1} \psi$$

$$= (\partial_\alpha + g^{-1} \partial_\alpha g + g L_\alpha g^{-1}) \psi$$

$$= (\partial_\alpha - L_\alpha^{(g)}) \psi$$

ψ solves eom $\Leftrightarrow L_\alpha^{(g)}$ takes the standard form

Suppose R_i is a projector, $R_i^2 = R_i$, $R_i^\dagger = R_i$.

$$g^{-1}(\lambda) = 1 - \sum_{i=1} \frac{R_i}{\lambda - \bar{\lambda}_i} = g(\bar{\lambda})^\dagger$$

Then

$$g(\lambda) g^{-1}(\lambda) = 1$$

$L_\alpha(\lambda)$ and $L_\alpha^{(g)}(\lambda)$ should be analytic except $\lambda=0, \infty$

But $L_\alpha^{(g)}(\lambda)$ has a simple pole at $\lambda = \bar{\lambda}_i$:

$$\text{Res}_{\lambda = \bar{\lambda}_i} L_\alpha^{(g)} = 0 \quad \text{for all } i \Rightarrow \text{fixes } R_i$$

Take $\lambda \rightarrow 0$ in the dressed connection:

$$\begin{aligned} L_x^{(g)} &= \frac{-i}{4\lambda} e^{\frac{i\phi^{(g)}}{2} \sigma_1} \sigma_3 e^{-\frac{i\phi^{(g)}}{2} \sigma_1} \\ &= \frac{-i}{4\lambda} g \sigma_3 g^{-1} - \underbrace{g^{-1} \partial_x g}_{\text{regular}} \end{aligned}$$

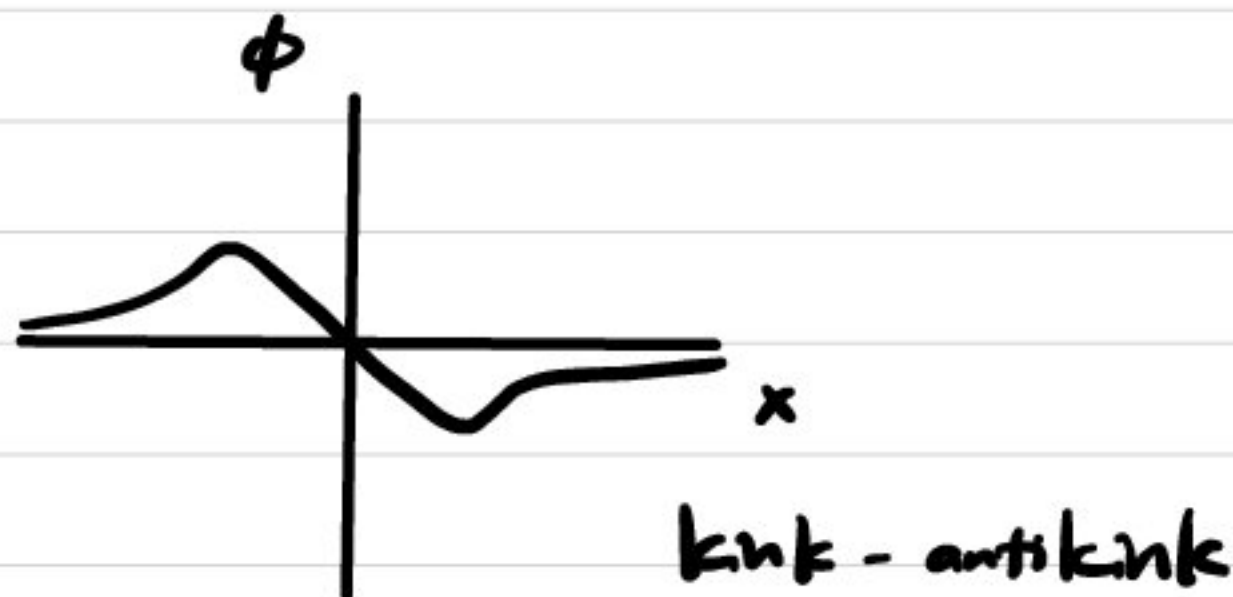
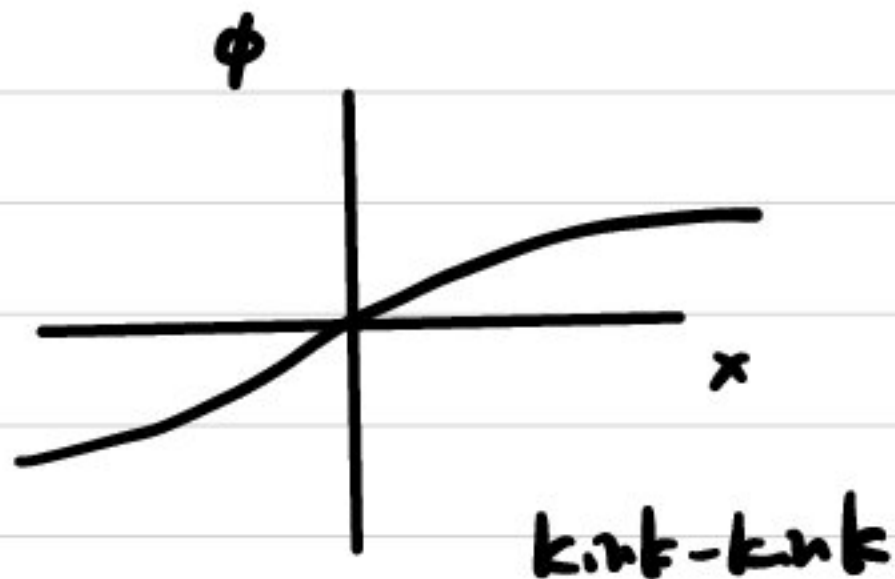
$$\Rightarrow g(\lambda) = \begin{pmatrix} \cos \phi/2 & i \sin \phi/2 \\ i \sin \phi/2 & \cos \phi/2 \end{pmatrix} + \mathcal{O}(\lambda) = 1 + \sum_i \frac{R_i}{\lambda - \lambda_i}$$

We can recover $\tan\left(\frac{\phi}{4}\right)$ from R_i 's.

Kink-kink solution $\left(\lambda_k = i a_k = i \sqrt{\frac{1-u_k}{1+u_k}} \right)$

$$\tan \frac{\phi}{4} = \frac{a_1 + a_2}{a_1 - a_2} \frac{e^{\xi_1} - e^{\xi_2}}{1 + e^{\xi_1 + \xi_2}}$$

kink-antikink solution : $\xi_2 \rightarrow \xi_2 + i\pi$



center-of-mass frame: $a_1 = 1/a_2 = \sqrt{\frac{1-v_1}{1+v_1}}$

$$X = \frac{x+t}{2}, \quad T = \frac{x-t}{2}, \quad \xi_1 = \gamma(x-vt) = \frac{x-vt}{\sqrt{1-v^2}}$$

$$\tan \frac{\phi}{4} = \frac{1}{v} \frac{\sinh(\gamma vt)}{\cosh(\gamma x)} \quad ; \quad \text{take } x, t \rightarrow \pm \infty$$

$$\rightarrow \frac{1}{v} \exp\left(\mp \frac{x-vt}{\sqrt{1-v^2}}\right) \equiv \exp\left(\mp \frac{x-v(t+\Delta T/2)}{\sqrt{1-v^2}}\right)$$

$$\Delta T = \frac{2}{\gamma v} \log v : \quad \text{time delay in CoM}$$

In the original giant magnon solution in $\mathbb{R}_t \times S^3$

$$E - J = \frac{\sqrt{\lambda}}{\pi} \sin \frac{P}{2}, \quad v = \frac{\partial E}{\partial P} = \cos \frac{P}{2}$$

Thus

$$\xi = \frac{x - (\cos \frac{P}{2}) t}{\sin \frac{P}{2}} \equiv \cosh \theta x - \sinh \theta t; \text{rapidity}$$

$$\tanh \theta = \cos \frac{P}{2}, \quad \frac{1}{\sinh \theta} = \gamma v$$

Time delay of particle 1 passing through particle 2 is

$$\Delta T_{12} = \sinh \theta_1 \log \tanh^2 \left(\frac{\theta_1 - \theta_2}{2} \right)$$

Consider "wave-function" for particle 1

$$\psi \sim e^{-iEt + ip(E)x - i\delta(E)}$$

$$= e^{-iE(t+\Delta T) + ip(E)x} \equiv e^{i\Phi(E)}$$

$$\frac{\partial \delta}{\partial E} = t + \Delta T - \frac{\partial p}{\partial E} x = \underbrace{\left(t - \frac{x}{v}\right)}_{\text{free particle phase}} + \Delta T$$

free particle phase

Thus phase shift is $\delta_{12} = \int dE, \Delta T_{12}$

AFS phase :

$$\theta = \frac{\sqrt{\lambda}}{2\pi} \sum_{r=2}^{\infty} \left\{ q_r(p_1) q_{r+1}(p_2) - q_{r+1}(p_1) q_r(p_2) \right\}$$

$$q_{r+1}(p) = \frac{i}{r} \left\{ \frac{1}{(x^+)^r} - \frac{1}{(x^-)^r} \right\}$$

$$\left\{ \begin{array}{ll} \text{in XXX model} & x^{\pm} \sim u \pm \frac{i}{2} \quad (\lambda \ll 1) \\ \text{in string theory} & x^{\pm} = \exp(\pm i p/2) \quad (\lambda \gg 1) \end{array} \right.$$

$$\text{sum} \rightarrow \theta(p_1, p_2) = \frac{\sqrt{\lambda}}{\pi} \left(\cos \frac{p_1}{2} - \cos \frac{p_2}{2} \right) \log(\dots)$$

Substitute p -variable:

$$\Delta T_{12} = \tan \frac{p_1}{2} \cdot \log \left(\frac{1 - \cos \frac{p_1 - p_2}{2}}{1 + \cos \frac{p_1 - p_2}{2}} \right)$$

time delay is related to the phase shift by

$$S_{12} = \int dE_1 \Delta T_{12} = \frac{\sqrt{\lambda}}{2\pi} \int dp_1 \sin \frac{p_1}{2} \log(\dots)$$

$$= -\frac{\sqrt{\lambda}}{\pi} \left\{ \underbrace{\left(\cos \frac{p_1}{2} - \cos \frac{p_2}{2} \right) \log(\dots)}_{\text{AFS phase}} + \underbrace{p_1 \sin \frac{p_2}{2}}_{\sim p_1 \epsilon_2, \text{ gauge choice}} \right\}$$

AFS phase

$\sim p_1 \epsilon_2$, gauge choice

We computed scattering phase from 2-magnon sol
& agrees with AFS phase

We don't get "matrix structure" of S -matrix
because we started from $\mathbb{R}_t \times S^2 \subset \text{AdS}_5 \times S^5$

Global charges $(\cancel{E}, \underbrace{S_1, S_2}, \underbrace{\cancel{J_1}, \cancel{J_2}, J_3})$
residual $\text{psu}(2|2)^2$ symmetry
fixes the matrix structure

Giant magnon at $p \rightarrow 0$

$\leadsto$ BMN or point-particle limit, $t = \phi = \tau$

Expand $AdS_5 \times S^5$ action around this 'vacuum'

light cone gauge: $X^0 = t, X^9 = \phi$

$X^i = y^i / g^{1/4} (i=1 \sim 8)$: physical

$$\mathcal{L} = \sqrt{\lambda} G_{MN}(X) \partial X^M \bar{\partial} X^N$$

$$= \partial y^i \bar{\partial} y^i + \frac{1}{\sqrt{\lambda}} \underline{V_4(y^i, y^i)} + \dots$$

Worldsheet 4pt also gives "S-matrix" in BMN limit

§ Superstring on $AdS_5 \times S^5$

[GSW textbook, Berkovits lectures, Arutyunov Frolov 0901.4937]

Bosonic:
$$L = \left(\sqrt{-\gamma} \gamma^{ab} G_{MN} + \epsilon^{ab} B_{MN} \right) \partial_a X^M \partial_b X^N$$

 $a, b = 0, 1$

Superstring $\left\{ \begin{array}{l} \text{worldsheet SUSY : } RNS \quad X^M, \psi^M \\ M = 0, 1, \dots, 9 \\ \text{spacetime SUSY : } GS, PS \\ X^M, \theta^A \\ A = 1, 2, \dots, 16 \end{array} \right.$

In RNS, Bosonization $\leadsto$ Spacetime fermions

$$e^{-\psi}$$

need to introduce 'picture' and PCO

D-brane background (as in AdS/CFT)

RR flux $\leadsto$ Non-zero torsion

not Levi-Civita connection, common in SUGRA

Supersymmetrize : $\Pi_a^\mu = \partial_a X^\mu - \frac{1}{2} \partial_a \theta^A (\Gamma^\mu)_{AB} \theta^B$

but θ^A has too many d.o.f.

Green-Schwarz flat space action

$$S_{GS} = S_{kin} + S_{WZ}$$

$$S_{kin} = -\frac{1}{2\pi} \int d^2\sigma \sqrt{-\gamma} \gamma^{ab} \pi_a^M \cdot \pi_b^M$$

$$S_{WZ} = -\frac{i}{2\pi} \int d^2z \epsilon^{ab} \partial_a X^M \cdot \theta^A \phi_{AB} \theta^B$$

(this is heterotic, needs (θ_1^A, θ_2^A) for type II)

$S_{WZ} \sim$ "fermionic B-field"

S_{GS} is invariant under global & local SUSY

local SUSY $\sim$ K-symmetry $\sim \sqrt{\text{Viracoro}}$

$$\begin{cases} \delta \theta^A = (\Gamma^M)^A_B \pi_a^M K^{Aa} \\ \delta X^M = i \theta^A (\Gamma^M)^A_B \delta \theta^B \end{cases}$$

Then $\delta \pi_a^M = -\partial_a \theta^A (\Gamma_a)_AB K^{Bb}$

$$\delta S_{\text{kin}} = -\frac{1}{\pi} \int \sqrt{-h} h^{ab} \pi_a^M \delta \pi_a^M$$

The second term was added such that

$$\delta S_{WZ} = -\frac{1}{\pi} \int \epsilon^{ab} \partial_a \theta^A (\mathcal{K}_b)_{AB} K^{Bb}$$

(using Fierz identity if needed)

Define the projector :

$$P_{\pm} \equiv \frac{1}{2} \left(h^{ab} \pm \frac{\epsilon^{ab}}{\Gamma - h} \right), \quad P_{\pm}^2 = P_{\pm}, \quad P_{\pm} P_{\mp} = 0$$

which can be written as $P_{\pm} = \frac{1}{2} (1 \pm \Gamma)$

We gauge-fix $SO(1,9)$ rotation s.t.

$$P_+ K^1 = K^1, \quad P_- K^2 = K^2$$

$$\text{Now } \delta S_{GS} = -\frac{2}{\pi} \int \sqrt{-h} \, \partial_\alpha \theta \, \underbrace{\mathcal{H} P_+ K^1}_{=0 \text{ for } K^2}$$

Thus half of θ^A go away by K -symmetry

Also $\delta^2 X \propto (\mathcal{H})^2 = 0$ by Virasoro constraints

$$\Rightarrow (K\text{-symmetry})^2 = \text{Virasoro}$$

GS action on supercoset is simple

$$\text{flat 10d : } \frac{\text{superpoincaré } (P, Q, M)}{SO(1,9) \quad (M)}$$

$$\text{AdS}_5 \times S^5 : \frac{PSU(2,2|4)}{SO(1,4) \times SO(5)}$$

$$\mathbb{Z}_4 \text{ grading : } g \in PSU(2,2|4), \quad j = g^{-1}dg$$

split j by the eigenspace of Ω

$$\Omega^4 = 1 \Rightarrow \text{eigenvalues } \{+1, i, -1, -i\}$$

An example:

$$\mathfrak{su}(4) = \text{Span} \left\{ T \in \text{Mat}(4, \mathbb{C}) \mid T^\dagger = -T, \text{tr}(T) = 0 \right\}$$

$$\dim_{\mathbb{R}} \mathfrak{su}(4) = 15$$

$$\mathfrak{so}(5) = \text{Span} \left\{ T \in \sqrt{-1} \text{Mat}(4, \mathbb{R}) \mid T^T = -T \right\}$$

$$\dim_{\mathbb{R}} \mathfrak{so}(5) = 10$$

$$\mathfrak{su}(4) = \underbrace{\text{"symmetric"}}_{\mathfrak{su}(4)} \oplus \underbrace{\text{"anti symmetric"}}_{\mathfrak{so}(5)}$$

$$\mathfrak{psu}(2,2|4) \subset \mathfrak{sl}(4|4); \quad M = \begin{pmatrix} B_1 & F_2 \\ F_3 & B_4 \end{pmatrix} \begin{matrix} \} 4 \\ \} 4 \end{matrix}$$

$\underbrace{\hspace{1.5cm}}_4 \quad \underbrace{\hspace{1.5cm}}_4$

$$K = \begin{pmatrix} 0 & -1 & & \\ 1 & 0 & & \\ & & 0 & -1 \\ & & 1 & 0 \end{pmatrix}, \quad K = \text{diag}(K, K)$$

$$\Omega(M) = -K M^{\text{st}} K, \quad M^{\text{st}} = \begin{pmatrix} B_1^t & -F_3^t \\ F_2^t & B_4^t \end{pmatrix}$$

$$\Omega(J_k) = i k J_k : \quad k=0,1,2,3, \quad \begin{array}{ll} \dim J_0 = 20 & (\text{gauge away}) \\ \dim J_2 = 10 & (\text{bosons}) \\ \dim J_1 = \dim J_3 = 16 & (\text{fermions}) \end{array}$$

Lagrangian for $AdS_5 \times S^5$ Green-Schwarz

$$\mathcal{L} = -\frac{\sqrt{\lambda}}{4\pi} \text{str} \left(\sqrt{-h} h^{\alpha\beta} \underbrace{J_\alpha^{(2)} J_\beta^{(2)}}_{\text{Bosonic kinetic}} + K \epsilon^{\alpha\beta} \underbrace{J_\alpha^{(1)} J_\beta^{(3)}}_{\text{Wess-Zumino term}} \right)$$

Bosonic kinetic

Wess-Zumino term

but bosonic $B_{MN} = 0$

- $\text{str}(J + c \mathbb{1}) = \text{str}(J)$

choose c s.t. $\text{tr}(J^{(2)}) = 0 \rightarrow$ "P" in $psu(2,2|4)$

- $K = \pm 1$ to have K -symmetry
kill half of fermionic d.o.f.

$$\begin{cases} P_- J^{(1)} = 0 \\ P_+ J^{(3)} = 0 \end{cases}$$

Lax connections

BPR form : $J = \frac{(1-z^2)^2}{2z^2} J^{(2)} - \frac{1}{2} \left(z^2 - \frac{1}{z^2} \right) * J^{(2)}$

[0305116]

$$+ (z-1) J^{(1)} + \left(\frac{1}{z} - 1 \right) J^{(3)}$$

- gauge transformation : $J' = h L h^{-1} + dh \cdot h^{-1}$

introduces z -independent terms including $J^{(0)}$

- Keep K -symmetry & flatness without eom : [0910.0221]

$$J_{\text{strong}} = J_{\text{BPR}} + \frac{1-z^4}{2} \left(C^{(0)} + \frac{C^{(1)}}{z^3} + \frac{C^{(3)}}{z} \right)$$

Coset coordinate

$$G_T = \exp(\chi) \exp(X)$$

$$X = \begin{pmatrix} 0 & Y \\ Y^\dagger & 0 \\ & 0 & iz \\ & iz^\dagger & 0 \end{pmatrix}, \quad \chi = \begin{pmatrix} & & \theta_1 & \theta_2 \\ & & \theta_3^\dagger & \theta_4^\dagger \\ -\theta_1^\dagger & \theta_3 & & \\ -\theta_2^\dagger & \theta_4 & & \end{pmatrix}$$

using $su(2,2) \oplus su(4) \supset su(2)^{\oplus 4}$

$$Z = \begin{pmatrix} z^{12} & -z^{11} \\ z^{22} & -z^{21} \end{pmatrix}, \quad Y = \begin{pmatrix} y^{34} & -y^{33} \\ y^{44} & -y^{43} \end{pmatrix}$$

If we use $z^{ai} \rightarrow (z^1, z^2, z^3, z^4)$

relation to the embedding coordinates of $S^5 \subset \mathbb{R}^6$

$$X^{2k-1} + \sqrt{-1} X^{2k} = \frac{z^{2k-1} + \sqrt{-1} z^{2k}}{1 + z^2/4}, \quad X^{5+i6} = \frac{1 - z^2/4}{1 + z^2/4} e^{i\phi}$$

Similarly for $AdS_5 \subset \mathbb{R}^{4,2}$

Metric:

$$ds^2 = - \left(\frac{1 + y^2/4}{1 - y^2/4} \right) dt^2 + \left(\frac{1 - z^2/4}{1 + z^2/4} \right) d\phi^2 + \frac{(dz^i)^2}{(1 + z^2/4)^2} + \frac{(dy^i)^2}{(1 - y^2/4)^2}$$

→ easy to expand around $y^i = z^i = 0$.